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Started	Wednesday, 21 October 2026, 1:26 PM
Completed	Wednesday, 21 October 2026, 1:33 PM
Duration	7 mins 2 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Quick Sort
b. Insertion Sort
c. Bubble Sort
d. Merge Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Binary Search

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting helps in finding duplicates quickly
b. Sorting makes it harder to search for items
c. Sorting is rarely used in programming
d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **4**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It is less efficient than bubble sort
c. It is based on the divide-and-conquer approach
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Quick sort
c. Merge sort
d. Heap sort

The correct answer is:
Merge sort

Question **6**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data in a single pass
c. Dividing the input into parts, solving each part, and combining the solutions
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **7**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Bubble Sort
- b. Binary Search
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **8**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Dividing the list into halves
- c. Bubbling up the largest element to its correct position
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **9**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sorted()
- b. order()
- c. sort()
- d. arrange()

The correct answer is: sorted()

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Provides a basis for other algorithms to work efficiently
c. Makes data harder to manage
d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is sorted independently
b. Each part is searched for a specific element
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **12**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It sorts data out of the box efficiently
b. It only works with integer arrays
c. It is less efficient than custom sorting algorithms
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

a. A sorting algorithm is stable if it preserves the order of all keys

b. A sorting algorithm is stable if it preserves the order of non-duplicate keys

c. A sorting algorithm is stable if it preserves the order of duplicate keys

d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Linear Search

b. Binary Search

c. Bubble Sort

d. Merge Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Complexity

c. Searching

d. Merging

The correct answer is:
Complexity